

CodeCook

AI-Powered Coding Practice Platform

Project Information:

- **Project Name:** CodeCook
 - **Project Type:** Full Stack Web Application
 - **Domain:** EdTech / Competitive Programming
 - **Developer:** Nithishwar T
 - **GitHub Repository:**
<https://github.com/nithishwar17/Coding-Platform>
 - **Live Demo:**
<https://coding-platform-6kv2.vercel.app/dashboard>
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1. Project Overview

CodeCook is a modern coding practice platform designed to help programmers improve their algorithmic thinking and problem-solving skills through an interactive coding environment.

The platform combines:

- Coding Challenges
- Online Code Editor
- AI Coding Mentor
- User Dashboard
- Submission Tracking
- Progress Analytics

into a single unified learning ecosystem.

The objective is to provide a LeetCode-style learning experience enhanced with Artificial Intelligence and modern web technologies.

2. Problem Statement

Many students struggle to find a platform that provides:

- Practice problems
- Interactive coding environment
- Learning assistance
- Progress tracking

- Personalized guidance

in one place.

CodeCook addresses this challenge by integrating coding practice, AI mentorship, and performance tracking into a single platform.

3. Project Objectives

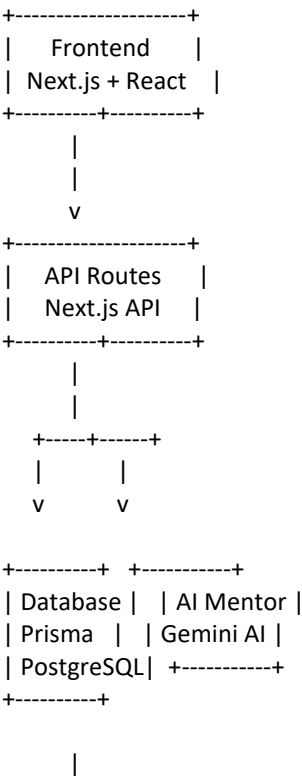
Primary Objectives

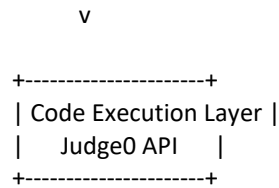
- Create an online coding practice platform.
- Allow users to solve programming problems.
- Track learning progress.
- Provide AI-powered mentoring.
- Build a scalable and responsive system.

Secondary Objectives

- Improve problem-solving skills.
- Provide coding interview preparation.
- Create a competitive programming ecosystem.

4. System Architecture





This architecture follows a serverless deployment model using Vercel and integrates external services for code execution and AI assistance.

5. Technology Stack

Frontend

- Next.js 14
- React.js
- TypeScript
- CSS Modules
- Monaco Editor

Backend

- Next.js API Routes
- Node.js

Database

- PostgreSQL
- Prisma ORM

Authentication

- NextAuth.js
- Google OAuth

Artificial Intelligence

- Google Gemini API

Deployment

- Vercel

Version Control

- Git
- GitHub

6. Core Modules

6.1 Authentication Module

Features:

- User Registration
- Login
- Google OAuth
- Session Management

6.2 Problem Management Module

Features:

- Coding Problem Listing
- Difficulty Levels
 - Easy
 - Medium
 - Hard
- Detailed Problem Descriptions
- Test Cases

6.3 Coding Playground

Features:

- Monaco Code Editor
- Language Selection
- Run Code
- Submit Solution

6.4 AI Mentor Module

Features:

- Coding Guidance
- Hints
- Problem Explanation

- Debugging Suggestions
- Markdown-formatted Responses

The AI mentor is configured to act as a teacher rather than directly providing complete solutions.

6.5 Dashboard Module

Features:

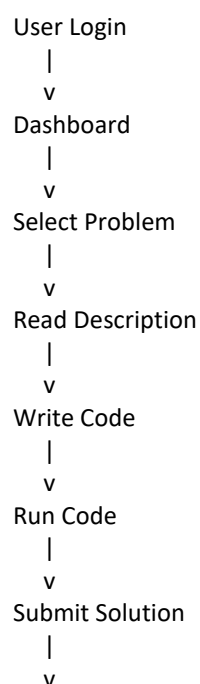
- Solved Problem Statistics
 - Progress Tracking
 - Submission Heatmap
 - Difficulty Distribution
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6.6 Submission Module

Features:

- Submission History
 - Runtime Status
 - Accepted / Wrong Answer Tracking
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7. Project Workflow / Flow Diagram



Code Execution
(Judge0 API)

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Result Evaluation

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Store Submission

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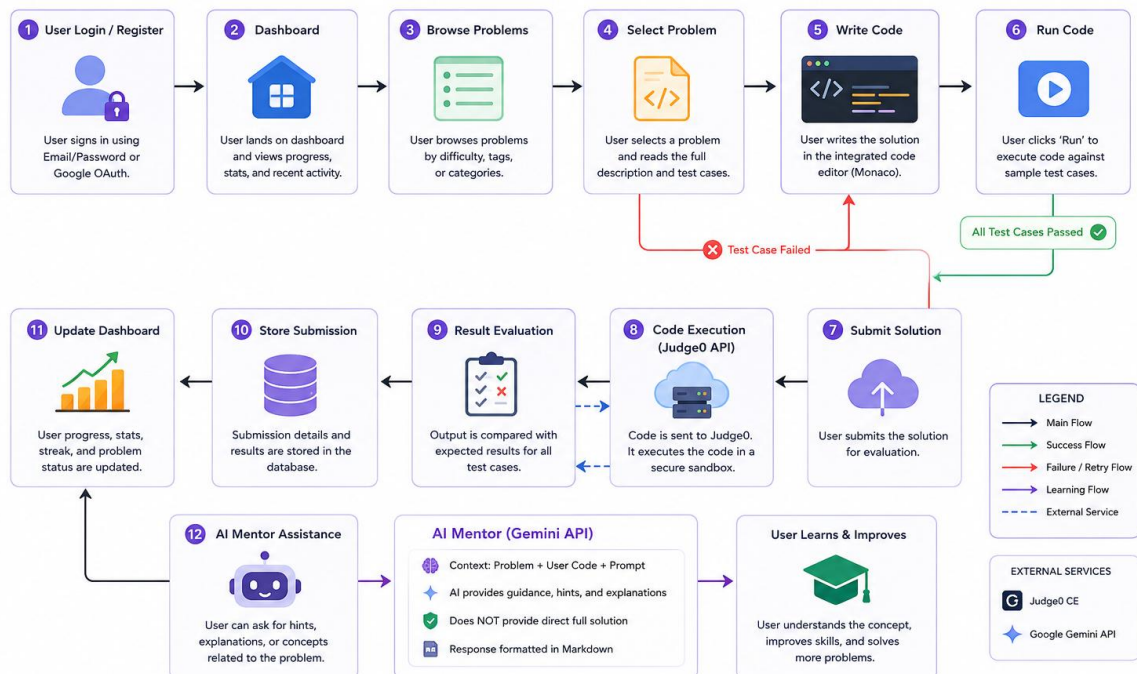
Update Dashboard

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AI Mentor Assistance

CODECOOK – PLATFORM WORKFLOW

End-to-End Flow of User Interaction and System Process



8. Database Design

User Table

Stores:

- User ID
- Email
- XP
- Streak
- Last Solved Date

Problem Table

Stores:

- Problem ID
- Title
- Difficulty
- Description
- Test Cases

Submission Table

Stores:

- User ID
 - Problem ID
 - Source Code
 - Language
 - Status
 - Submission Date
-

9. Key Features Implemented

User Authentication

- Secure login system
- Google OAuth support

Coding Environment

- Interactive Monaco Editor
- Multi-language support

AI Mentor

- AI-generated hints
- Learning assistance
- Markdown rendering

Dashboard Analytics

- Progress tracking
- Submission statistics

Responsive Design

- Dark mode interface
- Mobile-friendly layout

10. Challenges Faced

OAuth Redirect URI Issues

Problem:

- Google OAuth login failures.

Solution:

- Configured correct redirect URIs.
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Git Branch Management

Problem:

- Repository contained both main and master branches.

Solution:

- Migrated fully to main branch and removed master.
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Vercel Deployment Errors

Problem:

- TypeScript build failures.

Solution:

- Fixed missing field definitions and redeployed successfully.
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AI Model Availability

Problem:

- Gemini API service interruptions.

Solution:

- Added fallback model routing mechanism.

11. Testing

Functional Testing

- Login Functionality
- Problem Loading
- Code Submission
- AI Mentor Responses

Integration Testing

- Database Integration
- Authentication Integration
- AI Integration

Deployment Testing

- Vercel Production Deployment
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12. Future Enhancements

Short-Term

- Judge0 Integration Completion
- Additional Programming Languages
- Hidden Test Cases
- Better Submission Analytics

Long-Term

- Coding Contests
 - Global Leaderboards
 - Discussion Forums
 - AI Code Review
 - Personalized Learning Paths
 - Enterprise Interview Platform
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13. Project Achievements

- Successfully built and deployed a full-stack coding platform.
- Integrated AI-powered mentoring.
- Implemented secure authentication.

- Developed responsive coding workspace.
 - Created submission tracking system.
 - Deployed application on Vercel with GitHub CI/CD integration.
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14. Conclusion

CodeCook demonstrates the successful integration of modern web development, cloud deployment, AI assistance, and coding education into a single platform. The project showcases full-stack development skills, database management, authentication systems, AI integration, deployment practices, and software engineering principles, making it a strong final-year B.Tech project.

References

GitHub Repository

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Live Demo

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Technologies

- Next.js
- React.js
- Prisma ORM
- PostgreSQL
- NextAuth.js
- Google Gemini API
- Vercel
- GitHub